

# COLLECTING DATA & SCOPE OF INFERENCE

new 1.10–1.13 • old 3.1–3.5 • ★ old 3.6

## ZONE A • HOW THE DATA ARE COLLECTED

### Data from random sample or randomized experiment

|                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>SAMPLING METHODS</b><br><b>SRS:</b> every possible sample of size $n$ has the same chance.<br><b>Stratified:</b> divide into similar groups; randomly sample within each.<br><b>Cluster:</b> randomly choose whole groups; include everyone in them.<br><b>Systematic:</b> start randomly, then select every $k$ th individual. | <b>SOURCES OF BIAS</b><br><b>Undercoverage:</b> some groups are left out or hard to reach.<br><b>Nonresponse:</b> selected people do not respond.<br><b>Response bias:</b> wording or reporting distorts answers.<br><b>Voluntary response:</b> people choose to join the sample.<br><b>Convenience:</b> choose people who are easiest to reach. | <b>EXPERIMENTS</b><br><b>Control:</b> hold other conditions alike; compare treatments.<br><b>Random assignment:</b> use chance to form treatment groups.<br><b>Replication:</b> use enough experimental units.<br><b>Completely randomized:</b> assign all units at random.<br><b>Randomized block:</b> group similar units, then randomize within.<br><b>Matched pairs:</b> compare two related measurements or units.<br><b>Placebo:</b> an inactive treatment.<br><b>Blinding:</b> hide treatment identity from subjects or evaluators.<br><b>Confounding:</b> effects are mixed and cannot be separated. |
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## ZONE B • WHAT THE WORDS MEAN

|                                                                                                          |                                                                                                                     |                                                                                                                      |
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| <b>THE STUDY</b><br><b>Observational:</b> record what happens.<br><b>Experiment:</b> impose a treatment. | <b>THE PEOPLE OR OBJECTS</b><br><b>Population:</b> the full group of interest.<br><b>Sample:</b> the group studied. | <b>THE MEASUREMENTS</b><br><b>Treatment:</b> a condition imposed.<br><b>Response variable:</b> the outcome measured. |
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## ZONE C • WHAT CAN WE CONCLUDE?

|                    | Random assignment: YES                   | Random assignment: NO                   |
|--------------------|------------------------------------------|-----------------------------------------|
| Random sample: YES | Generalize: yes<br>Cause and effect: yes | Generalize: yes<br>Cause and effect: no |
| Random sample: NO  | Generalize: no<br>Cause and effect: yes  | Generalize: no<br>Cause and effect: no  |

## ZONE D • WATCH OUT + BEYOND

Random **selection** lets you generalize to the population.  
Random **assignment** supports cause and effect.

★ **BEYOND THE EXAM**  
Re-randomize groups many times. If the observed difference is rare, it is evidence against chance alone.